

## EXERCISE - 6

### **CloudSim Custom Scheduling Using Shortest Job First (SJF)**

#### **Aim:**

To implement and simulate a **custom Shortest Job First (SJF) scheduling algorithm in CloudSim** by extending the default `DatacenterBroker`, and to verify that Cloudlets with shorter lengths are completed before longer Cloudlets.

#### **Algorithm:**

1. Start the Ubuntu virtual machine.
2. Install the required prerequisites: **Java JDK, Git, Maven, and IntelliJ IDEA Community Edition**.
3. Clone the CloudSim repository from GitHub.
4. Open the cloned CloudSim project in IntelliJ IDEA as a Maven project and allow the required dependencies to load.
5. Create a custom broker class named `SJFDatacenterBroker` by extending the CloudSim `DatacenterBroker`.
6. Override the `submitCloudlets()` method in the custom broker.
7. Sort the available Cloudlets in ascending order according to their **Cloudlet length**, so that the shortest job is given priority.
8. Assign the sorted Cloudlets to the available virtual machines and invoke the normal CloudSim Cloudlet submission process.
9. Create the `SJFSimulation` main class and initialize the CloudSim environment.
10. Create a Datacenter, Host, custom SJF broker, and Virtual Machine.
11. Create three Cloudlets with different lengths: **40000, 10000, and 20000**.

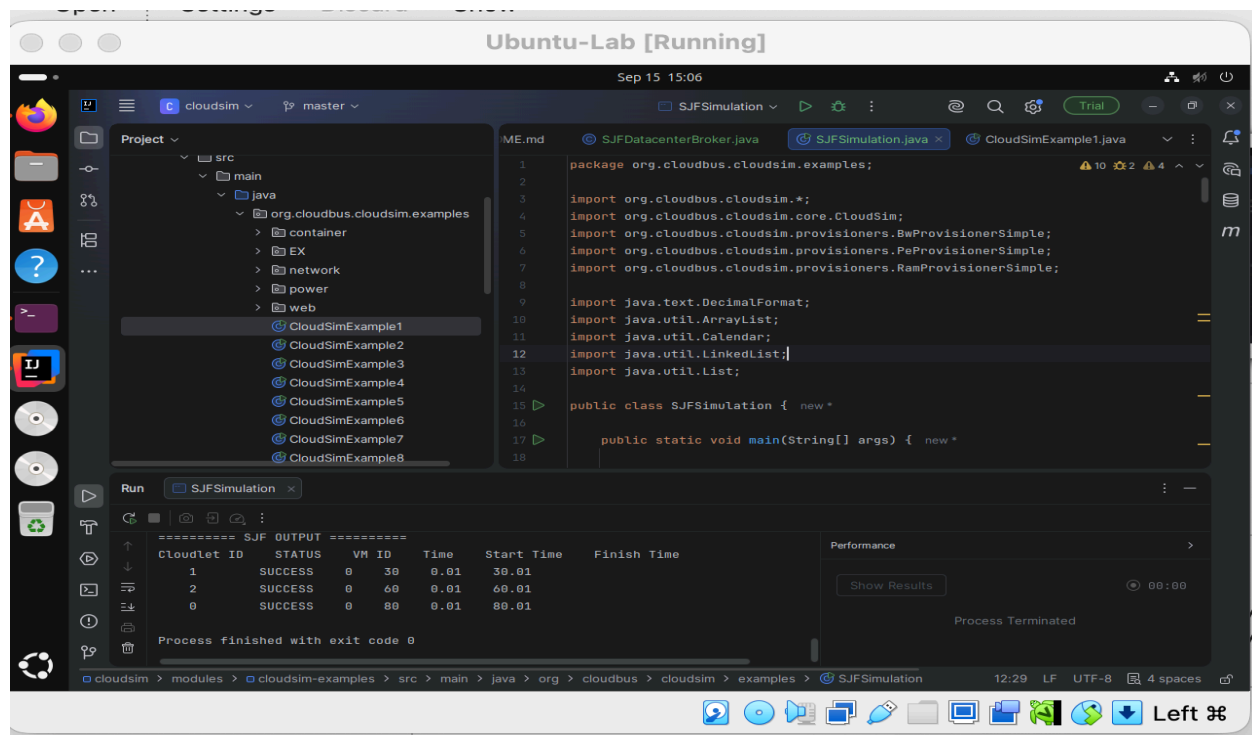
12. Submit the Cloudlets to the custom SJF broker and start the simulation.
13. Collect the completed Cloudlets and display their status, VM ID, execution time, start time, and finish time.
14. Verify that the shortest Cloudlet completes first, followed by the medium and longest Cloudlets.
15. Test each menu option and exit the application successfully.

## Output:

Therefore, the observed execution/completion order was:

**Cloudlet 1 → Cloudlet 2 → Cloudlet 0**

This matches the expected SJF order specified in the experiment.



## Result:

Thus, the **Shortest Job First (SJF) scheduling algorithm was successfully implemented and simulated using CloudSim**. The custom broker prioritized Cloudlets based on their job length, and the simulation verified that the shortest Cloudlet completed first, followed by the medium and longest Cloudlets.